

Tobias Gabriel Young

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OBJECTIVE

Mechanical engineering student at Cornell University with hands-on experience in CAD design, machining, and software development. Proven ability to contribute to high-impact engineering teams through work with Cornell Hyperloop and industry partners. Seeking opportunities to apply skills in mechanical design, fabrication, and problem-solving to real-world engineering challenges.

EDUCATION

Cornell University, College of Engineering

Bachelor of Science in Mechanical and Aerospace Engineering · Ithaca, NY · Expected May

2028 GPA: 3.77

Relevant Coursework: Statics and Mechanics of Solids (A-), Thermodynamics (B+), Physics: Electromagnetism (A), Physics: Mechanics and Heat (A), Differential Equations (A), Multivariable Calculus (B), Engineering General Chemistry (A)

SKILLS

Beginner: ANSYS, MATLAB

Intermediate: AutoCAD, Java, Lathe, Python, SOLIDWORKS

Advanced: Bandsaw, Belt Sander, Bridgeport Mill, Conflict Resolution, Fusion 360, Microsoft Applications

Certifications: Financial Literacy, Grassroots Certified, Red Apron Certification

Other: 98 WPM Typing Speed

WORK EXPERIENCE

Cornell Hyperloop – Mechanical Engineer

September 2025 – Present · Ithaca, NY

- Designed and fabricated a pneumatic braking system for a magnetic-levitation pod using SolidWorks, machining, and testing
- Designed an initial propulsion system using a brushless DC motor, gearbox, and wheels to achieve liftoff prior to LIM engagement
- Communicated with industry partners, including Maxon Motors, to define design requirements and support a \$20,000 sponsorship

Curious Cardinals – Mentor

April 2024 – Present · Remote

- Mentored students in coding, teaching logic, sequencing, and problem-solving via personalized projects using Python on Replit
- Designed interactive lessons that simplified technical concepts and fostered an early interest in programming and creativity
- Communicated with parents and students to establish achievable goals, provide feedback, and adjust instruction accordingly

Conti Civil LLC – Project Engineer Intern

May 2024 · Newark, NJ

- Inspected and evaluated daily construction job site progress for the Hudson Tunnel Project to ensure accuracy •
- Negotiated with suppliers to coordinate and schedule loam, soil, and silt deliveries, ensuring appropriate quantities
- Attended daily morning coordination meetings to review progress across 5+ active work zones

Eastern Football Club – Coach

September 2022 – September 2024 · Briarcliff Manor, NY

- Led dynamic training sessions for boys and girls using the 'play, practice, play' structure to build core skills •
- Adapted to challenging situations between peers, parents, and players, managing diverse groups of 20–30 players •
- Fostered clear communication with parents and players to strengthen teamwork, accountability, and development

PROJECTS

SLF Density Separator

MAE2250, Team LanternBee · Cornell University

- Designed a sugar water-based separation system to remove Spotted Lanternfly from mechanically harvested grapes at point of collection
- Nominated for Outstanding Problem/Solution Award — selected as 1 of 10 finalist teams out of 50, placed 6th
- Built and tested a functional prototype using laser-cut acrylic, PVC pipe, and perforated aluminum sheet

Frictional Brakes for Maglev Pod

SolidWorks · December 2025

- Designed a dual-clamp pneumatic braking system using a double-acting actuator and spring-assisted C-bracket for balanced braking
- Modeled the braking assembly in SolidWorks and evaluated performance and stability in ANSYS for each design iteration
- Selected actuators, springs, and bolts while coordinating mechanical, pneumatic, and control constraints with a dead man's switch

Tic Tac Toe with Event Driven Logic

Python · December 2025

- Developed an interactive Tic Tac Toe game using event-driven programming with the Turtle graphics library •
- Implemented game-state management and win-condition logic using NumPy arrays, including turn validation and board resets
- Designed a rule-based computer opponent capable of winning, blocking player moves, and adapting strategy based on game state

Snake Game with Dual Food Logic

Python · May 2025

- Developed a snake game with multi-food logic, score tracking, and linear interpolation for animated graphics •
- Integrated custom game design into tutoring sessions to explain advanced coding concepts through interactive exercises

Solubility Prediction Using Linear Regression

Python · May 2025

- Built a linear regression model to predict compound solubility from molecular data using scikit-learn, pandas, and Python
- Generalized machine learning code across multiple datasets, ensuring reproducibility and reliable results

EXTRACURRICULAR ACTIVITIES

Cornell Mundial Soccer Club

August 2024 – Present

- Goalkeeper for Cornell's premier club soccer team, demonstrating discipline and decision-making under high pressure
- Balanced academic coursework with travel, training, and intercollegiate matches

Sigma Pi Fraternity – Steward

April 2025 – Present

- Serve on the executive board, contributing to chapter governance and decision-making
- Created and manage the chapter meal plan contract, coordinating food service logistics and vendor expectations
- Facilitate communication between Gill Grilling, the Sigma Pi Alumni Board, and the undergraduate chapter • Address and resolve conflicts and house issues on a near-weekly basis, mediating between members and stakeholders

ACHIEVEMENTS & CERTIFICATIONS

- Outstanding Problem/Solution Award Nominee — MAE2250, Cornell University, 2025 (Top 10 of 50 teams)
- Financial Literacy Certification
- Grassroots Certified — Coaching certification
- Red Apron Certification

INTERESTS

Soccer, Weightlifting, Mountain Biking, Guitar, Clarinet, Oil Painting

REFERENCES

Available upon request.